HOSPITAL MANAGEMENT SYSTEM

```
DROP DATABASE IF EXISTS HospitalManagementSystem;
CREATE DATABASE HospitalManagementSystem;
USE HospitalManagementSystem;

CREATE TABLE Department (
  department_id INT PRIMARY KEY AUTO_INCREMENT,
  department_name VARCHAR(100) NOT NULL UNIQUE,
  location VARCHAR(100) DEFAULT 'Main Building',
  phone VARCHAR(15) UNIQUE
);

CREATE TABLE Doctor (
  doctor_id INT PRIMARY KEY AUTO_INCREMENT,
  doctor_name VARCHAR(100) NOT NULL,
  specialization VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE,
  phone VARCHAR(15) UNIQUE,
  salary DECIMAL(10,2) CHECK (salary > 0),
  department_id INT NOT NULL,
  FOREIGN KEY (department_id) REFERENCES Department(department_id)
);

CREATE TABLE Patient (
  patient_id INT PRIMARY KEY AUTO_INCREMENT,
  patient_name VARCHAR(100) NOT NULL,
  age INT CHECK (age BETWEEN 0 AND 120),
  gender VARCHAR(10) CHECK (gender IN ('Male','Female','Other')),
  phone VARCHAR(15) UNIQUE,
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  email VARCHAR(100) UNIQUE,
  address VARCHAR(200),
  blood_group VARCHAR(5) DEFAULT 'Unknown'
);

CREATE TABLE Room (
  room_id INT PRIMARY KEY AUTO_INCREMENT,
  room_number VARCHAR(10) NOT NULL UNIQUE,
  room_type VARCHAR(30) NOT NULL
    CHECK (room_type IN ('General','Private','ICU')),
  charge_per_day DECIMAL(10,2) NOT NULL CHECK (charge_per_day > 0),
  room_status VARCHAR(20) DEFAULT 'Available'
    CHECK (room_status IN ('Available','Occupied'))
);

CREATE TABLE Appointment (
  appointment_id INT PRIMARY KEY AUTO_INCREMENT,
  patient_id INT NOT NULL,
  doctor_id INT NOT NULL,
  appointment_date DATE NOT NULL,
  appointment_time TIME NOT NULL,
  reason VARCHAR(255),
  status VARCHAR(20) DEFAULT 'Scheduled'
    CHECK (status IN ('Scheduled','Completed','Cancelled')),
  FOREIGN KEY (patient_id) REFERENCES Patient(patient_id),
  FOREIGN KEY (doctor_id) REFERENCES Doctor(doctor_id)
);
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CREATE TABLE Admission (
   admission_id INT PRIMARY KEY AUTO_INCREMENT,
   patient_id INT NOT NULL,
   room_id INT NOT NULL,
   admission_date DATE NOT NULL,
   discharge_date DATE,
   diagnosis VARCHAR(255) NOT NULL,
   CHECK (discharge_date IS NULL OR discharge_date >= admission_date),
   FOREIGN KEY (patient_id) REFERENCES Patient(patient_id),
   FOREIGN KEY (room_id) REFERENCES Room(room_id)
);

CREATE TABLE Prescription (
   prescription_id INT PRIMARY KEY AUTO_INCREMENT,
   appointment_id INT NOT NULL,
   medicine_name VARCHAR(100) NOT NULL,
   dosage VARCHAR(100) NOT NULL,
   duration_days INT CHECK (duration_days > 0),
   FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
);

CREATE TABLE Bill (
   bill_id INT PRIMARY KEY AUTO_INCREMENT,
   patient_id INT NOT NULL,
   bill_date DATE NOT NULL,
   amount DECIMAL(10,2) CHECK (amount >= 0),
   payment_status VARCHAR(20) DEFAULT 'Pending'
      CHECK (payment_status IN ('Pending','Paid')),
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   FOREIGN KEY (patient_id) REFERENCES Patient(patient_id)
);

INSERT INTO Department (department_name,location,phone) VALUES
('Cardiology','Block A','9876500011'),
('Neurology','Block B','9876500012'),
('Orthopedics','Block C','9876500013'),
('Pediatrics','Block D','9876500014'),
('General Medicine','Block E','9876500015');

INSERT INTO Doctor
(doctor_name,specialization,email,phone,salary,department_id) VALUES
('Dr. Ravi Kumar','Cardiologist','ravi@hospital.com','9000000001',120000,1),
('Dr. Priya Sharma','Neurologist','priya@hospital.com','9000000002',130000,2),
('Dr. Arjun Rao','Orthopedic Surgeon','arjun@hospital.com','9000000003',110000,3),
('Dr. Sneha Reddy','Pediatrician','sneha@hospital.com','9000000004',100000,4),
('Dr. Kiran Kumar','Physician','kiran@hospital.com','9000000005',90000,5);

INSERT INTO Patient
(patient_name,age,gender,phone,email,address,blood_group) VALUES
('Rahul',25,'Male','8000000001','rahul@gmail.com','Vijayawada','O+'),
('Anjali',30,'Female','8000000002','anjali@gmail.com','Guntur','A+'),
('Suresh',45,'Male','8000000003','suresh@gmail.com','Eluru','B+'),
('Lakshmi',60,'Female','8000000004','lakshmi@gmail.com','Vijayawada','AB+'),
('Kiran',12,'Male','8000000005','kiran@gmail.com','Machilipatnam','O+');

INSERT INTO Room
(room_number,room_type,charge_per_day,room_status) VALUES
```

('G101','General',1500,'Available'),

('G102','General',1500,'Available'),

('P201','Private',3500,'Available'),

('P202','Private',3500,'Occupied'),

('I301','ICU',8000,'Occupied');

INSERT INTO Appointment

(patient_id,doctor_id,appointment_date,appointment_time,reason,status) VALUES

(1,1,'2026-09-05','10:00:00','Chest Pain','Scheduled'),

(2,2,'2026-09-05','11:00:00','Headache','Completed'),

(3,3,'2026-09-06','12:00:00','Knee Pain','Scheduled'),

(4,5,'2026-09-07','09:30:00','Fever','Completed'),

(5,4,'2026-09-07','10:30:00','Regular Checkup','Scheduled');

INSERT INTO Admission

(patient_id,room_id,admission_date,discharge_date,diagnosis) VALUES

(3,4,'2026-09-01',NULL,'Knee Injury'),

(4,5,'2026-09-02',NULL,'Heart Problem'),

(1,1,'2026-08-25','2026-08-28','Fever');

INSERT INTO Prescription

(appointment_id,medicine_name,dosage,duration_days) VALUES

(1,'Aspirin','1 tablet twice daily',5),

(2,'Paracetamol','1 tablet after food',3),

(3,'Ibuprofen','1 tablet twice daily',7),

(4,'Azithromycin','1 tablet once daily',5);

INSERT INTO Bill

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(patient_id,bill_date,amount,payment_status) VALUES
(1,'2026-09-01',5000,'Paid'),
(2,'2026-09-02',3500,'Paid'),
(3,'2026-09-03',12000,'Pending'),
(4,'2026-09-03',25000,'Pending'),
(5,'2026-09-04',2000,'Paid');

SELECT * FROM Patient;

UPDATE Patient
SET phone='8111111111',address='Vijayawada'
WHERE patient_id=1;

SELECT
a.appointment_id,p.patient_name,d.doctor_name,
d.specialization,a.appointment_date,a.appointment_time,a.status
FROM Appointment a
JOIN Patient p ON a.patient_id=p.patient_id
JOIN Doctor d ON a.doctor_id=d.doctor_id;

SELECT d.doctor_name,d.specialization,dp.department_name
FROM Doctor d
JOIN Department dp ON d.department_id=dp.department_id;

SELECT p.patient_name,a.diagnosis,r.room_number,r.room_type,r.charge_per_day
FROM Admission a
JOIN Patient p ON a.patient_id=p.patient_id
JOIN Room r ON a.room_id=r.room_id;
```

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SELECT d.doctor_name,d.specialization,a.appointment_id
FROM Doctor d
LEFT JOIN Appointment a ON d.doctor_id=a.doctor_id;

SELECT patient_name
FROM Patient
WHERE patient_id IN (
  SELECT patient_id FROM Bill
  WHERE amount > (SELECT AVG(amount) FROM Bill)
);

SELECT doctor_name,salary
FROM Doctor
WHERE salary > (SELECT AVG(salary) FROM Doctor);

SELECT patient_name
FROM Patient
WHERE patient_id = (
  SELECT patient_id FROM Bill
  ORDER BY amount DESC LIMIT 1
);

SELECT dp.department_name,COUNT(d.doctor_id) AS doctor_count
FROM Department dp
LEFT JOIN Doctor d ON dp.department_id=d.department_id
GROUP BY dp.department_id,dp.department_name;
```

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```
SELECT dp.department_name,COUNT(d.doctor_id) AS doctor_count
FROM Department dp
JOIN Doctor d ON dp.department_id=d.department_id
GROUP BY dp.department_id,dp.department_name
HAVING COUNT(d.doctor_id)>1;

CREATE VIEW AppointmentDetails AS
SELECT
a.appointment_id,p.patient_name,d.doctor_name,
d.specialization,dp.department_name,
a.appointment_date,a.appointment_time,a.reason,a.status
FROM Appointment a
JOIN Patient p ON a.patient_id=p.patient_id
JOIN Doctor d ON a.doctor_id=d.doctor_id
JOIN Department dp ON d.department_id=dp.department_id;

SELECT * FROM AppointmentDetails;

SELECT * FROM AppointmentDetails
WHERE status='Scheduled';

DELIMITER //

CREATE PROCEDURE GetPatientAppointments(IN p_patient_id INT)
BEGIN
SELECT
p.patient_name,d.doctor_name,d.specialization,
a.appointment_date,a.appointment_time,a.status
```

```
FROM Appointment a
JOIN Patient p ON a.patient_id=p.patient_id
JOIN Doctor d ON a.doctor_id=d.doctor_id
WHERE p.patient_id=p_patient_id;
END //

DELIMITER ;

CALL GetPatientAppointments(1);

SELECT * FROM Room
WHERE room_status='Available';

SELECT * FROM Room
WHERE room_type='ICU';

SELECT b.bill_id,p.patient_name,b.amount,b.bill_date
FROM Bill b
JOIN Patient p ON b.patient_id=p.patient_id
WHERE b.payment_status='Pending';

SELECT SUM(amount) AS total_revenue
FROM Bill
WHERE payment_status='Paid';

SELECT MAX(amount) AS highest_bill
FROM Bill;
```

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```
SELECT COUNT(*) AS total_patients
FROM Patient;


SELECT doctor_name,specialization
FROM Doctor
WHERE department_id=(
   SELECT department_id
   FROM Department
   WHERE department_name='Cardiology'
);
```